

# Test Diverse Users

Project: Transparent Book Recommender

## 1. Your mission

### Your learning target

I can evaluate across varied cases.

I can report failures with their denominators.

I can separate development evidence from final checks.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Week 34: Test Diverse Users

Evaluation must include ordinary, edge, and underserved cases.



## THINKER MISSION

Run four user profiles; log errors and fairness concerns.

Thinker name and date:

**Write or draw your reasoning and evidence.**

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## Readiness for the connected explanation

If final cases guide a code change, are they still untouched evaluation cases?

## This week's end goal

By the end, I can evaluate varied case types and prioritise an important failure slice.

# 2. Notice the evidence

## PICTURE STORY

Notice → Predict → Explain

Begin with visible evidence. Do not explain more than the picture and information support.



MYSTERY

The learners must use test diverse users to solve a problem. What information do they have? What decision or calculation do they make? What would count as evidence that it worked?

Three observations (what you can point to):

**Write or draw your reasoning and evidence.**

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My prediction and the clue supporting it:

**Write or draw your reasoning and evidence.**

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## A closer explanation

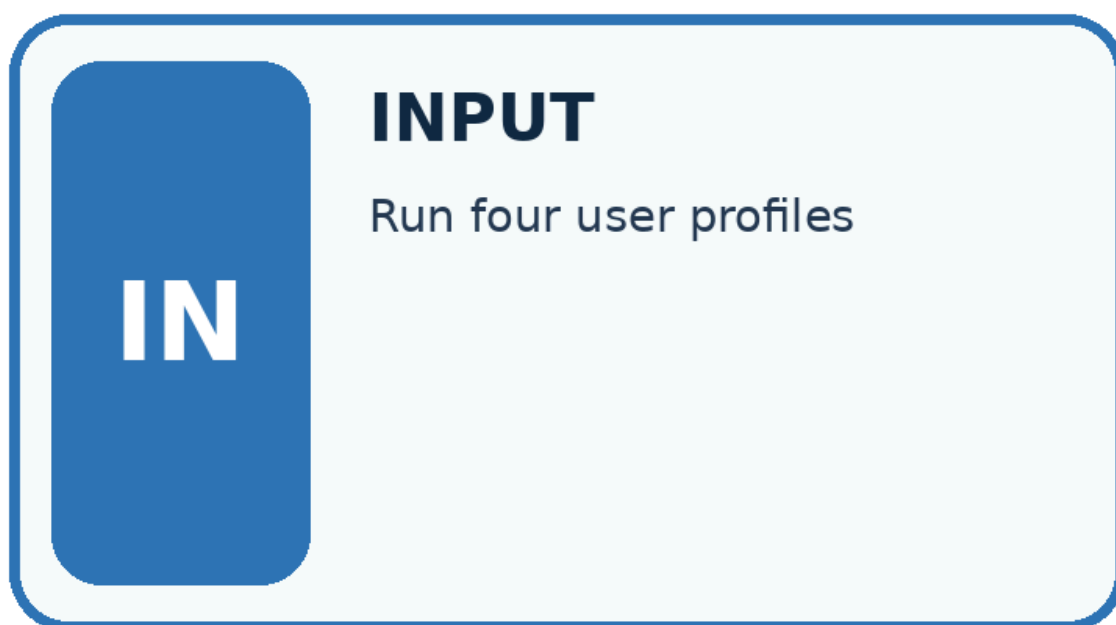
A fictional chooser passes 5 of 6 development checks but fails its only tie case. An overall pass rate can hide that concentrated weakness. Evaluate ordinary, boundary, invalid, empty and tie cases separately, using requirements rather than duplicated versions of one easy example.

## 3. See how it works

VISUAL MECHANISM

Input → Process → Output

Trace the information. At each stage, ask what changed and how you could test it.



↓



## PROCESS

Evaluation must include ordinary, edge, and underserved cases.





## OUTPUT

log errors and fairness concerns.

Which stage could fail even when the other two are correct? Explain:

**Write or draw your reasoning and evidence.**

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## A closer explanation

```
checks = [{"category": "ordinary", "passed": True},
          {"category": "ordinary", "passed": True},
          {"category": "boundary", "passed": True},
          {"category": "invalid", "passed": True},
          {"category": "empty", "passed": True},
          {"category": "tie", "passed": False}]

passed = 0

for check in checks:
    if check["passed"]:
        passed = passed + 1

print(passed, len(checks))
```

## Follow the evidence

### Step or case · Evidence · Meaning

**Step or case:** Overall

**Evidence:** 5 of 6

**Meaning:** One failure remains

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**Step or case:** Tie category

**Evidence:** 0 of 1

**Meaning:** Known weakness

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**Step or case:** Other categories

**Evidence:** Limited cases

**Meaning:** No universal guarantee

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Read each row from left to right and connect the evidence to the stated rule.

The script reports 5 and 6. About 83.3 percent of these checks pass, but that is not an accuracy estimate

for all future use. The cases were deliberately chosen, their frequencies are artificial, and each category has few cases.

### Look inside the overall result

The six development checks include ordinary, boundary, invalid, empty and tie cases. Five pass, but the only tie case fails. Saying five out of six pass is true; saying the tie rule works is false. Keep both statements visible. A category table helps you choose the next repair. Adding many copies of passing ordinary cases can raise the overall fraction without checking a new condition or fixing the failed requirement.

### Connect the picture and the explanation

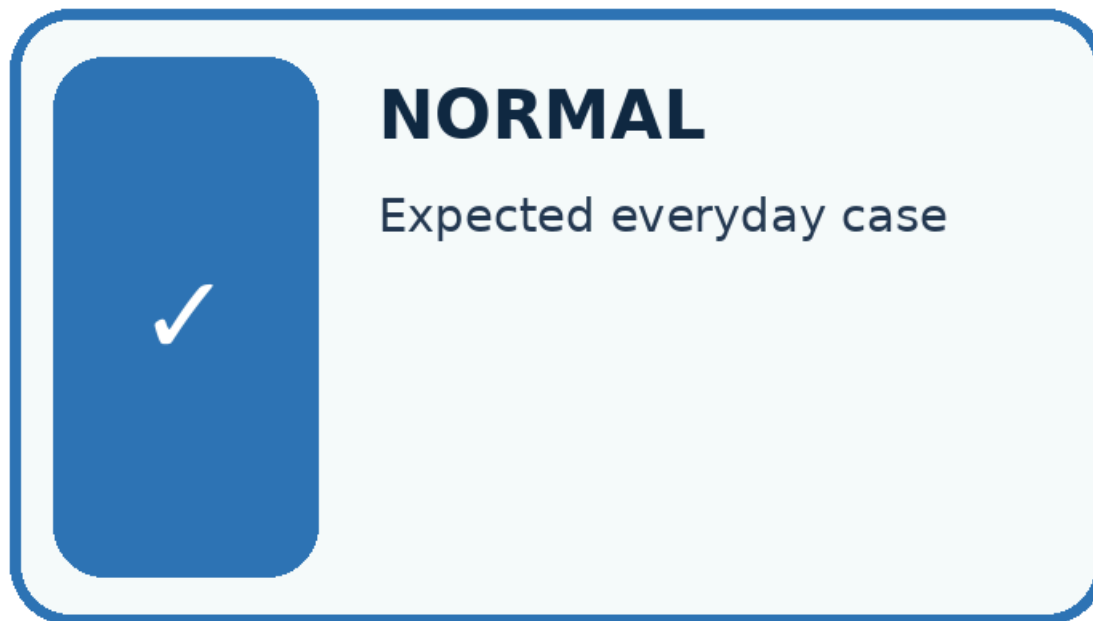
Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

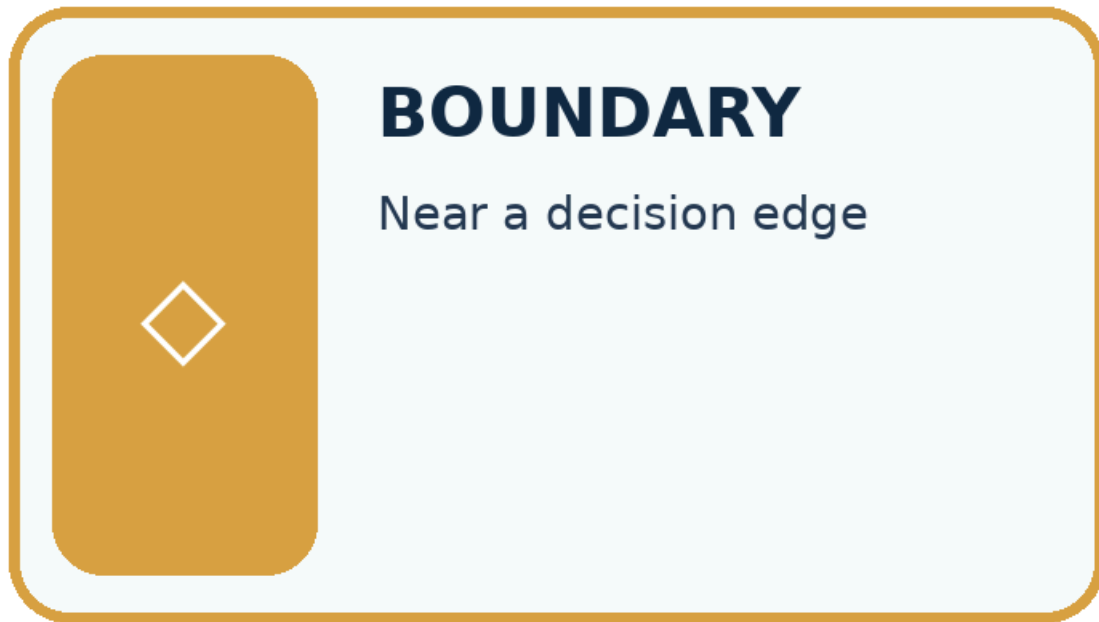
## 4. Compare the cases

### CASE GALLERY

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.

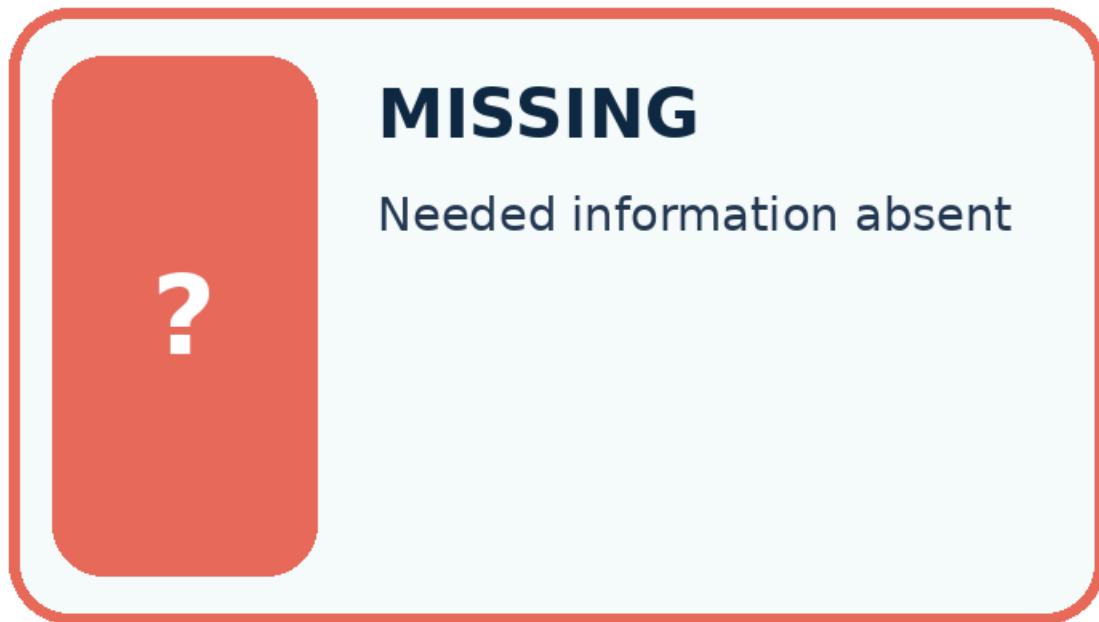




## **BOUNDARY**

Near a decision edge

NORMAL CASE: predict and name evidence. BOUNDARY CASE: predict and name evidence.



## **MISSING**

Needed information absent



MISSING INFORMATION: predict and name evidence. TRANSFER CASE: predict and name evidence.

Which case is most likely to reveal a hidden assumption? Why?

**Write or draw your reasoning and evidence.**

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### Change one thing in the pictured case

Change: Someone reports only the ordinary slice.

Prediction: The claim becomes 4/4 on ordinary cases, not 100% across all tests.

Reason: Eight nonordinary cases and their failures have been omitted.

## 5. Words that unlock the idea

### VOCABULARY LAB

#### Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

**1**

## **TEST PROFILE**

constructed user case for  
evaluation

**2**

## **EDGE CASE**

unusual but plausible input

3

## UNDERSERVED

receives weaker outcomes  
or coverage

4

## FAIRNESS CONCERN

evidence of uneven quality  
or access

**Word · My own example · Not this / boundary**

**Word:** test profile

**My own example:** A description of the kinds of cases and conditions included in an evaluation.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** edge case

**My own example:** An unusual, boundary or difficult input that can reveal a failure hidden by routine

examples.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** underserved

**My own example:** Receiving less adequate support or performance than the intended service requires.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

**Word:** fairness concern

**My own example:** A reason to examine whether a system creates unjustified differences in treatment or outcomes.

**Not this / boundary:** \_\_\_\_\_

\_\_\_\_\_

## Words used in the closer explanation

### Word · Meaning

**Word:** Evaluation set

**Meaning:** Cases used to assess a specified version

\_\_\_\_\_

**Word:** Failure category

**Meaning:** A type of scenario where an expected result is missed

\_\_\_\_\_

**Word:** Pass rate

**Meaning:** Passing cases divided by executed cases

\_\_\_\_\_

**Word:** Coverage

**Meaning:** The variety of requirements and conditions tested

\_\_\_\_\_

## 6. Follow a worked example

WORKED EVIDENCE

## Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete evidence.

### WEEK 34 • WORKED EVIDENCE

- 1 Profiles: typical reader; very short-book need; bilingual preference; screen-reader format need.
- 2 Run the same frozen recommender for all four.
- 3 Record expected, actual, score evidence, and issue type.
- 4 Do not average away one group's repeated failure.

1. Profiles: typical reader; very short-book need; bilingual preference; screen-reader format need.
2. Run the same frozen recommender for all four.
3. Record expected, actual, score evidence, and issue type.
4. Do not average away one group's repeated failure.

Recalculate, retrace, or restate the reasoning in your own representation:

**Write or draw your reasoning and evidence.**

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CHECK Did the evidence answer the exact question? Did the total, path, or source record stay consistent?

### A closer explanation

**If ten copies of one passing ordinary case are added, what becomes the pass fraction? Does coverage improve?**

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**Learn from the worked case**

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

**A pictorial worked case: Test Diverse Users**

A fictional recommender is tested on distinct ordinary, equality, invalid-input and tie cases.

**The facts we will use**

Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

Categories are disjoint and every case was run on the same frozen version with the same contract.

## Disjoint evaluation categories

|          | Full passes | Cases run |
|----------|-------------|-----------|
| Ordinary | 4           | 4         |
| Equality | 1           | 3         |
| Invalid  | 2           | 3         |
| Ties     | 1           | 2         |

Read a cell using both its row and column labels.

## Test Diverse Users

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Total count

Passes  $4+1+2+1=8$ ; cases  $4+3+3+2=12$ .



#### 2. Overall rate

$8/12 \approx 66.7\%$  full passes.



#### 3. Find weak slices

Equality  $1/3$ ; invalid  $2/3$ ; ties  $1/2$ ; ordinary  $4/4$ .



#### 4. Prioritise repair

Inspect failed equality traces and any invalid-input safety consequence, not only the total.

### CHANGE ONE THING

Someone reports only the ordinary slice.

Explain the changed result using the matching lesson.

## Read the picture one step at a time

### Step · What happens and why

**Step:** 1. Total count

**What happens and why:** Passes  $4+1+2+1=8$ ; cases  $4+3+3+2=12$ .

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**Step:** 2. Overall rate

**What happens and why:**  $8/12 \approx 66.7\%$  full passes.

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### Step: 3. Find weak slices

**What happens and why:** Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4.

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### Step: 4. Prioritise repair

**What happens and why:** Inspect failed equality traces and any invalid-input safety consequence, not only the total.

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Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

## Why this result follows

Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting. Prioritisation also depends on consequences: a malformed input that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected output.

## What this example does not establish

Twelve designed cases describe this test suite, not an unbiased estimate of future usage accuracy.

Read the labelled picture: Can these records establish performance for different user groups?

Change one thing: Someone reports only the ordinary slice. Predict the result before checking the explanation. What remains the same?

## 7. Find and repair a mistake

### ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak evidence.



### COMPARE • DIAGNOSE

Weak idea: Good average performance proves every user is served well. Strong move: locate evidence, change one cause, and retest.

**Evidence record · Expected · Actual · Likely cause / next test**

**Evidence record:** normal case

**Expected:** works

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** boundary case

**Expected:** stated rule

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** missing input

**Expected:** safe response

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

**Evidence record:** fresh transfer

**Expected:** generalizes

**Actual:** \_\_\_\_\_

**Likely cause / next test:** \_\_\_\_\_  
\_\_\_\_\_

Repair one cause. Why is this repair better than changing several things?

**Write or draw your reasoning and evidence.**

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## A closer explanation

For a rule-based chooser, evaluate conformance to its policy separately from whether users find suggestions useful. Neither a pass fraction nor an invented satisfaction score proves educational benefit.

**You fix the tie bug using the final tie case. What should the next evaluation report say?**

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## 8. Practise together

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



### TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET evidence\_record PREDICT expected\_result RUN one normal case and one boundary case  
COMPARE actual\_result with expected\_result IF different: locate first cause; change one factor; RETEST  
EXPLAIN result, limitation, and next safe step

Trace one input. Which line changes the state or chooses a path?

**Write or draw your reasoning and evidence.**

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### A closer explanation

Hiding the tie failure in an average conceals an unmet requirement. Preserve the failed input, expected result, actual result and version. If you repair using it, classify it as development/regression evidence and reserve new final cases.

### Practise with less help: Plan coverage

Use workbook W2 for the connected example “Evaluating varied cases”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

#### Plan · Do · Check

**Plan:** What is given? Which rule or procedure applies?

**Do:** Record each intermediate step.

**Check:** Does the result follow? What remains unknown?

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Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for Evaluating varied cases. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Can these records establish performance for different user groups?

**My explanation using a pictured step and the given facts**

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This is supported practice. Keep the separately named independent case for an attempt before feedback.

## 9. Run the investigation

### EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



### **PREDICT → TEST → RECORD → EXPLAIN**

Create four meaningfully different profiles, predict outcomes, run, classify failures, and compare results by profile rather than only overall.

Create four meaningfully different profiles, predict outcomes, run, classify failures, and compare results by profile rather than only overall.

**Test / input · Prediction · Actual · Evidence / calculation · Change or keep?**

**Test / input:** normal

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** boundary

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** missing

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

**Test / input:** fresh transfer

**Prediction:** \_\_\_\_\_

**Actual:** \_\_\_\_\_

**Evidence / calculation:** \_\_\_\_\_

**Change or keep?:** \_\_\_\_\_

\_\_\_\_\_

What did the hardest test teach you that the normal test did not?

**Write or draw your reasoning and evidence.**

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\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

**Plan → try → check**

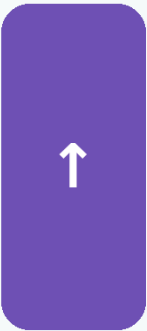
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

## 10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the worked example.



## MASTERY LADDER

1 Name it • 2 Trace it • 3 Apply it • 4 Challenge it • 5 Explain limits

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or evidence path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest evidence of independent understanding:

Write or draw your reasoning and evidence.

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# 11. Connect your project

## CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



## TRANSPARENT BOOK RECOMMENDER

Use the same test record across ordinary, boundary, missing-data, and access-needs cases.

**THIS WEEK'S CHECKPOINT** Use the same test record across ordinary, boundary, missing-data, and access-needs cases.

**SYSTEM MAP** Reader need → allowed inputs → transparent rule or score → ranked candidates → explanation → test record → human decision

What will you add, change, or verify in the project this week?

**Write or draw your reasoning and evidence.**

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What should remain under human control?

**Write or draw your reasoning and evidence.**

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**RESPONSIBLE DESIGN** Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

**Evidence for your project**

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition coverage.

## 12. Show what you know

### INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



**INDEPENDENT PROOF**

Run four user profiles; log errors and fairness concerns.

CHALLENGE Run four user profiles; log errors and fairness concerns.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

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2. My evidence that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

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3. One limitation, fairness concern, or situation needing human review:

**Write or draw your reasoning and evidence.**

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SELF-CHECK ☐ I explained the mechanism ☐ I used evidence ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

## 13. Study a complete case

### Week 34 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

#### The situation

A recommender satisfies  $\frac{5}{6}$  adventure requests and  $\frac{3}{6}$  poetry requests.

#### The question

Compare and investigate.

#### Worked explanation

Rates are  $\frac{5}{6}$  and  $\frac{3}{6}$ . Inspect catalogue coverage, labels and scoring; twelve tests do not establish cause.

#### Explain it in your own words

Which detail supports the answer, and why does it matter?

**Write or draw your reasoning and evidence.**

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Trace the supplied case visually

Compare success for different interests

**Evidence or stage · Value or result**

**Evidence or stage:** Adventure requests

**Value or result:** 5/6

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**Evidence or stage:** Poetry requests

**Value or result:** 3/6

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A recommender satisfies 5/6 adventure requests and 3/6 poetry requests.

Compare and investigate.

Rates are 5/6 and 3/6. Inspect catalogue coverage, labels and scoring; twelve tests do not establish cause.

## 14. Try a new case independently

**Independent case: Evaluate independently**

Given ordinary 3/3, boundary 1/2, invalid 2/2 and tie 0/1, compute overall results and prioritize issues.

**My independent explanation, trace and limits**

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case

already practised with feedback cannot establish fresh transfer.

**Week 34 Independent application**

Close the worked explanation. Apply the idea to this different question without copying.

A recommender serves 6 of 6 easy-book requests but 1 of 4 advanced-book requests. Report both results and identify the next test priority.

My answer, with a trace, calculation or supporting evidence

**Write or draw your reasoning and evidence.**

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**Create and test**

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

**Write or draw your reasoning and evidence.**

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**Connect to your project**

Use the same test record across ordinary, boundary, missing-data, and access-needs cases.

The evidence I will keep

**Write or draw your reasoning and evidence.**

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Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

### Transfer practice from the original programme

The recommender succeeds on 9/10 mystery requests and 2/5 poetry requests. Compute total and separate rates. A team reports only 11/15. Explain what that hides.

**Explain the mechanism, evidence and one limit.**

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### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

## 15. Your lesson route

Start with this question: A fictional recommender is tested on distinct ordinary, equality, invalid-input and tie cases.

**By the end, I can · How I will show it**

**By the end, I can:** Evaluate across varied cases

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Report failures with their denominators

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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**By the end, I can:** Separate development evidence from final checks

**How I will show it:** Explain the deciding step, show a trace or test, and state one limit.

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Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

## 16. Finish and return

Close your notes. Explain this goal in your own words: Evaluate across varied cases

**My deciding step, evidence and one remaining question**

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If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.

## 17. Supported practice, repair and portfolio

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional recommender is tested on distinct ordinary, equality, invalid-input and tie cases.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? Inspect failed equality traces and any invalid-input safety consequence, not only the total.

1. Prioritise repair

2. Total count
3. Find weak slices
4. Overall rate

**My selected conclusion, deciding fact and explanation**

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Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Prioritise repair Inspect failed equality traces and any invalid-input safety consequence, not only the total. Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting. Prioritisation also depends on consequences: a malformed input that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected output.

**Practice 2: explain the deciding evidence**

Changed-case practice: Someone reports only the ordinary slice. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The claim becomes 4/4 on ordinary cases, not 100% across all tests.
3. We can decide the result without examining the changed input or the procedure.

**My selected conclusion, deciding fact and explanation**

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Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must

say what follows from those facts.

After your attempt, compare with the supported explanation: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Eight nonordinary cases and their failures have been omitted.

## **Repair a missing building block**

Review Build the Recommender (ai-2/33). Goal: Implement a deterministic recommender. Short practice: Explain why B does not win after D is excluded even though it has fewer pages.. Return to this week and explain what you changed.

Review Permissions and Human Review (ai-2/31). Goal: Explain the mechanism and terminology of least privilege and tool scopes. Short practice: Which proposal is allowed, and why is an authorised object alone insufficient?. Return to this week and explain what you changed.

Review Prompt Experiments (ai-2/28). Goal: Explain the mechanism and terminology of controlled prompt experiments. Short practice: Name the case P passes and Q fails, and explain why it matters.. Return to this week and explain what you changed.

## **Predict, run and debug**

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

## **Keep your completed work**

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.